

English
Medium



EDU TERIA

72nd BPSC

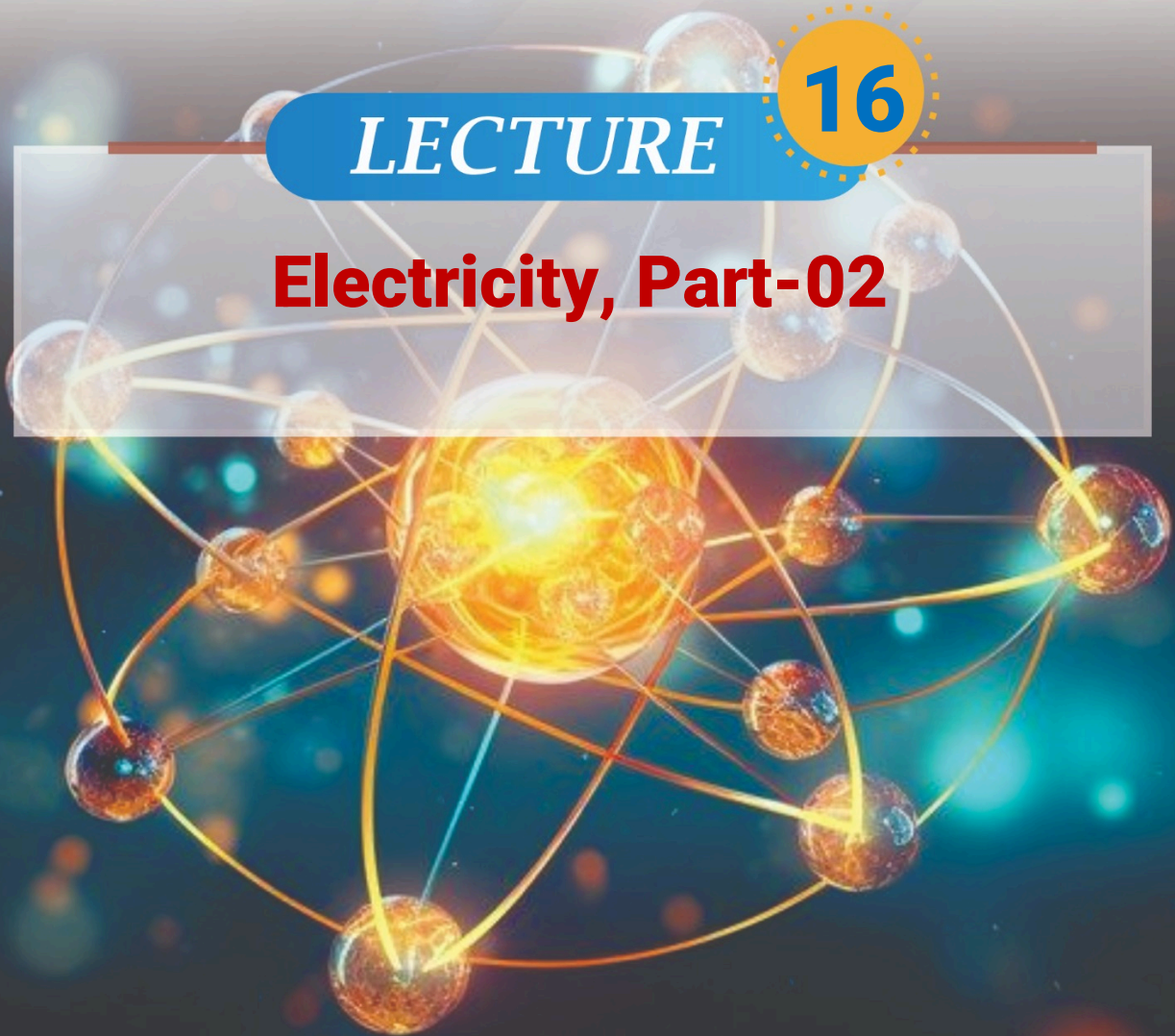
Prelims Online Batch

PHYSICS

LECTURE

16

Electricity, Part-02



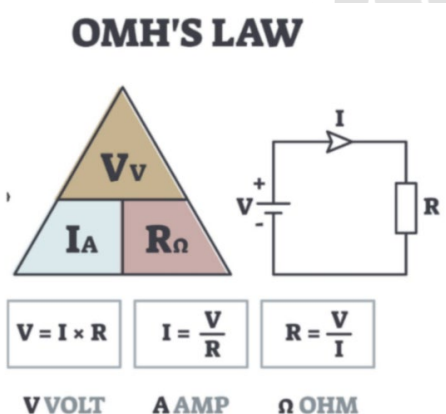
Electricity, Part-02

Ohm's Law

At constant temp, if a conducting wire with different potential connected than flow of electric current is directly proportional difference.

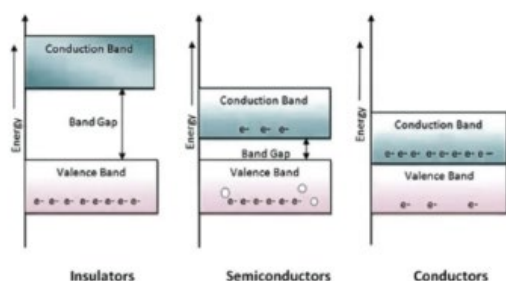
Relationship between V, I & R,

- $V = IR$
- Valid for ohmic conductors



Classification of material on the basis of conductivity

1. Conductor 2. Insulator 3. Semi-conductor

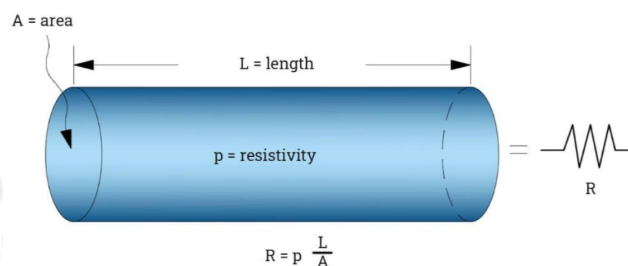


Few free electrons are present at room temperature.

Conductors	Insulator	Semi-conductors
• Free e^- are present hence can flow electric current through conductor. • On increasing temperature flow of electric current decrease.	• Free e^- are not available hence, electric current cannot pass through it. • Not available	• Few e^- are available at room temperature, hence can conduct electricity. • On increasing temp, free e^- increases, hence, flow of current increase.

Resistance (R)

Resistance is the property of a material that opposes the flow of electric current. It tells how difficult it is for electrons to move through a conductor



- **Unit:** Ohm (Ω)

1. Length (L)

- Longer wire more resistance
- $R \propto L$

2. Area (A)

- Thicker wire less resistance

$$R \propto \frac{1}{A}$$

3. Material (Resistivity p)

- Copper low resistance
- Rubber very high resistance

4. Temperature

- **Metals:** resistance increases with temperature
- **Semiconductors:** resistance decreases with temperature

$$R = \delta \frac{\ell}{A}$$

R = Resistance

δ = Resistivity

ℓ = Length of conductor

A = Cross Sectional area

Combination of Resistors

A combination of resistors refers to the connection of two or more resistors in a circuit in such a way, either in series, parallel, or a combination of both, which collectively provides a specific total resistance, helps control the flow of current, and distributes voltage

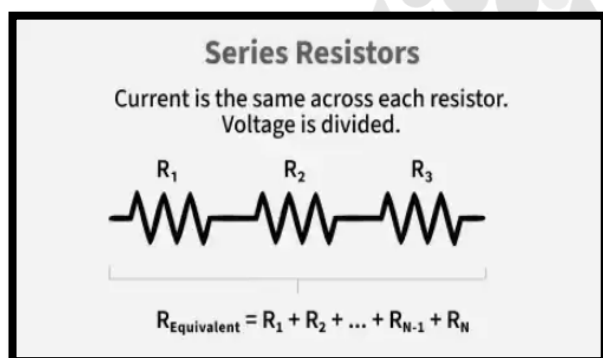




appropriately across different parts of the circuit for the proper functioning of electrical devices.

I. Series Combination:

When two or more resistors are connected one after another in a single path such that the current has only one path to flow, the arrangement is known as a series combination of resistors. In this type of connection, the same current flows through each resistor because charge leaving one resistor directly enters the next without any branching. This makes the current identical at every point in the circuit.



II. Parallel Combination

When two or more resistors are connected between the same two points such that there are multiple paths for current to flow, the arrangement is called a parallel combination of resistors. In this type of connection, the potential difference across each resistor remains the same, while the total current gets divided among different branches. The current entering the junction splits into different parts, say I_1 and I_2 , flowing through R_1 and R_2 respectively, and the total current is given by $I = I_1 + I_2$

